FH | JOANNEUM
Electronic Engineering

Testing of Automotive Systems (Part I)

Module 5 – CAN / LIN / FlexRay

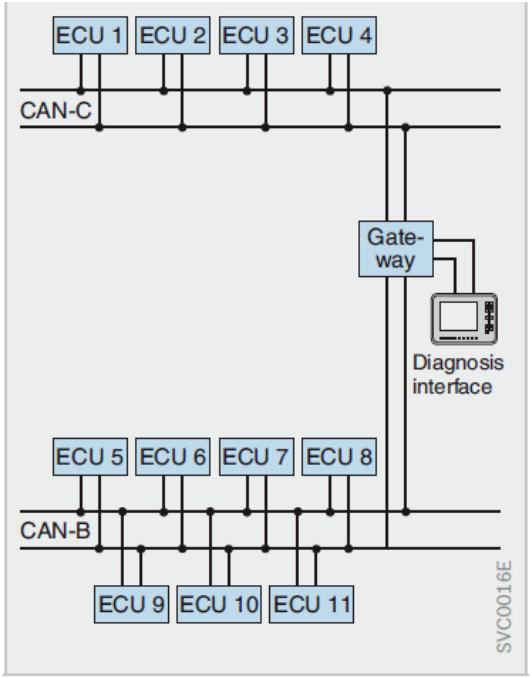
Martin Lamprecht, Magna Steyr

FH JOANNEUM
Electronic Engineering

CAN-BUS

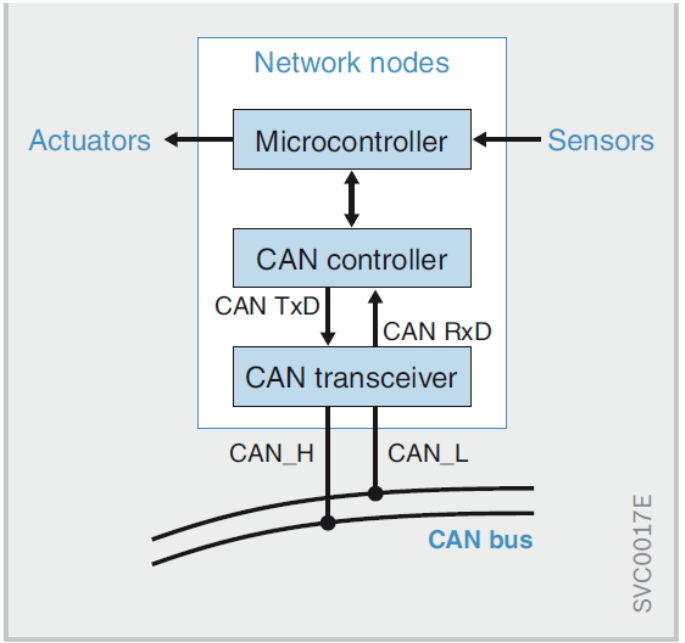
CAN-Bus

- Firstly introduced 1991, still standard in automotive
- Used in various domains in vehicles
- Different data rates are used
  - CAN-C 125kbit-1Mbit/sec
    - „high-speed CAN“ 500 kbit/sec
    - Still in almost every vehicle
  - CAN-B 5-125kbit/sec
    - „low-speed CAN“ 125 kbit/sec
    - More fault-tolerant
    - less used



Network nodes

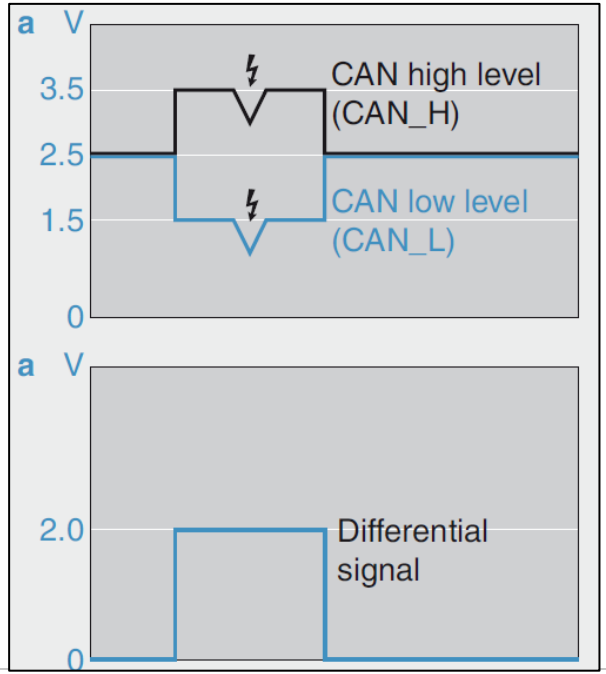
- All network nodes are connected to a bus and each node is able to receive all information sent on the bus
- The two bus lines are designated CAN_H and CAN_L
- A network node comprises of
  - Microcontroller
    - runs the application program
    - controls the CAN controller
    - prepares the sent data and evaluates received data
  - CAN Controller
    - responsible for the transmit and receive modes
    - generates the data communication bit stream
    - forwards it to the transceiver on the TxD line
  - CAN Tranceiver
    - Signal amplification, generates voltage levels
    - transmits the processed bit stream serially



FH JOANNEUM
Electronic Engineering

Logical bus states

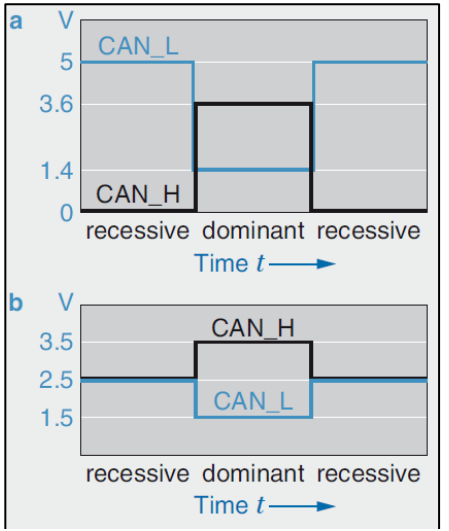
- Two states created by CAN controller
  - dominant binary “0”
  - recessive “1”
  - NRZ = non-return-to-zero
- Unshielded twisted pair cables (UTP, diameter between 0,34 and 0,6 mm²)
- Disturbance pulses have effect on both lines
- Differential amplifier subtracts CAN_L from CAN_H level
- Additional shielding reduce self emissions

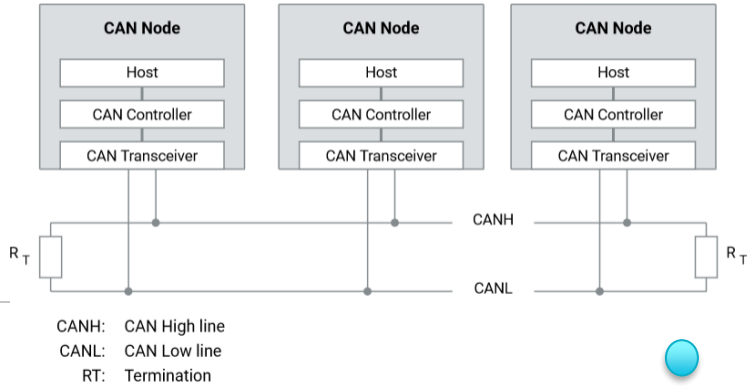


FH JOANNEUM
Electronic Engineering

Electrical bus states

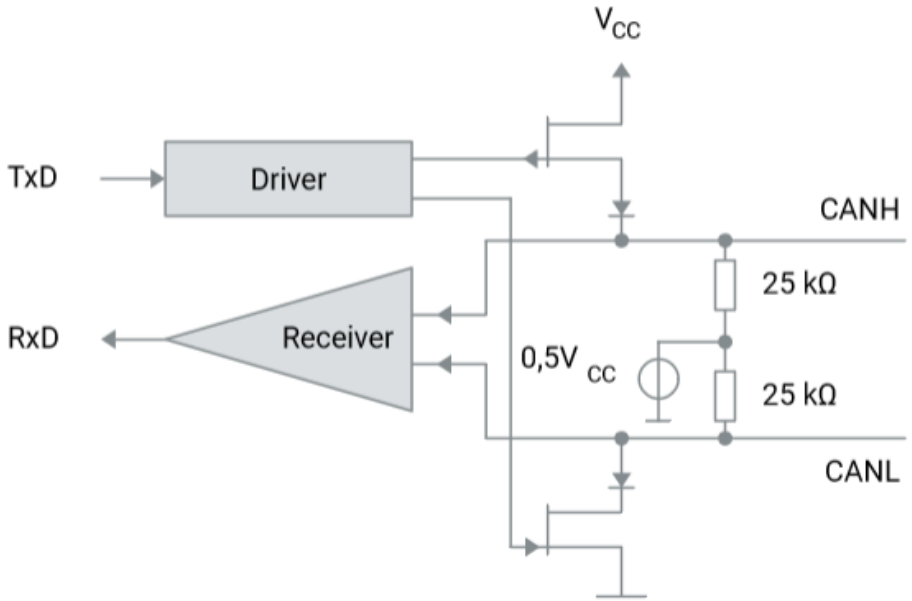
- CAN transceiver converts digital to voltage signals
  - Low-Speed CAN (a)
    - dominant CAN_H 3.6V / CAN_L 1.4V
    - recessive CAN_H 0V / CAN_L 5V
  - High-Speed CAN (b)
    - dominant CAN_H 3.5V / CAN_L 1.5V
    - recessive CAN_H 2.5V / CAN_L 2.5V
- Terminal resistor RT= 120Ω to dampen reflections (b)
- Maximum number of nodes 32 according ISO11898
- Maximum bus length (recommendation)
  - 500kbaud 100m
  - 125kbaud 500m

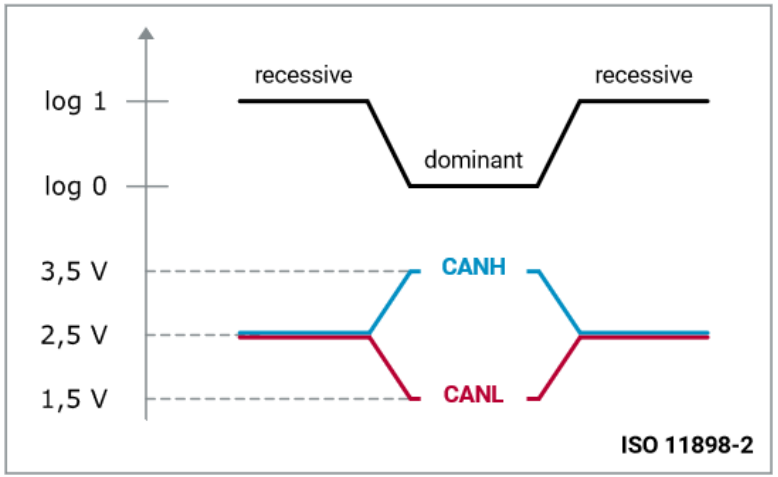




FH JOANNEUM
Electronic Engineering

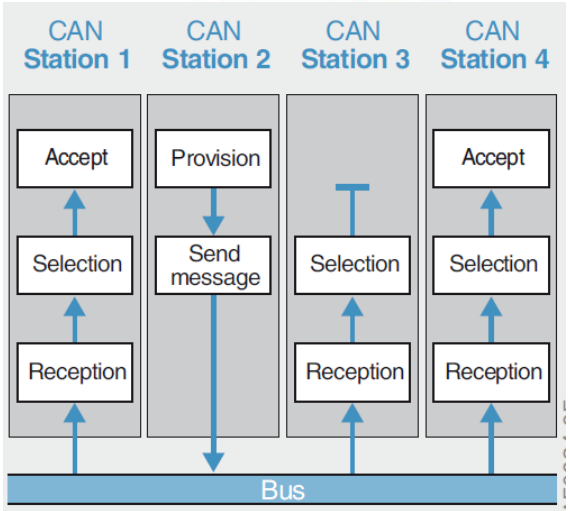
CAN Tranceiver

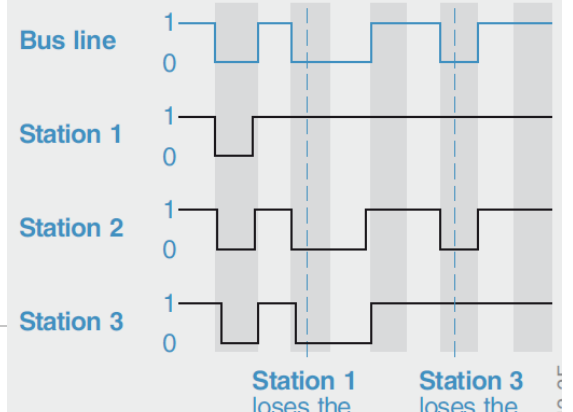




CAN Communication

- Multi-Master Principle
  - Each node may send messages any time
  - If bus is free and arbitration has been passed
- Content-based addressing
  - No addressing of nodes but of messages
  - Identifier classifies message content
  - Messages are broadcasted to all stations
  - Read only those messages which are stored in acceptance list
- Arbitration
  - Message begins with a dominant bit (start-of-frame bit), followed by the identifier
  - Message with highest priority is assigned first access
  - With highest priority bus access ~300µs
  - The higher the busload the bigger the latency

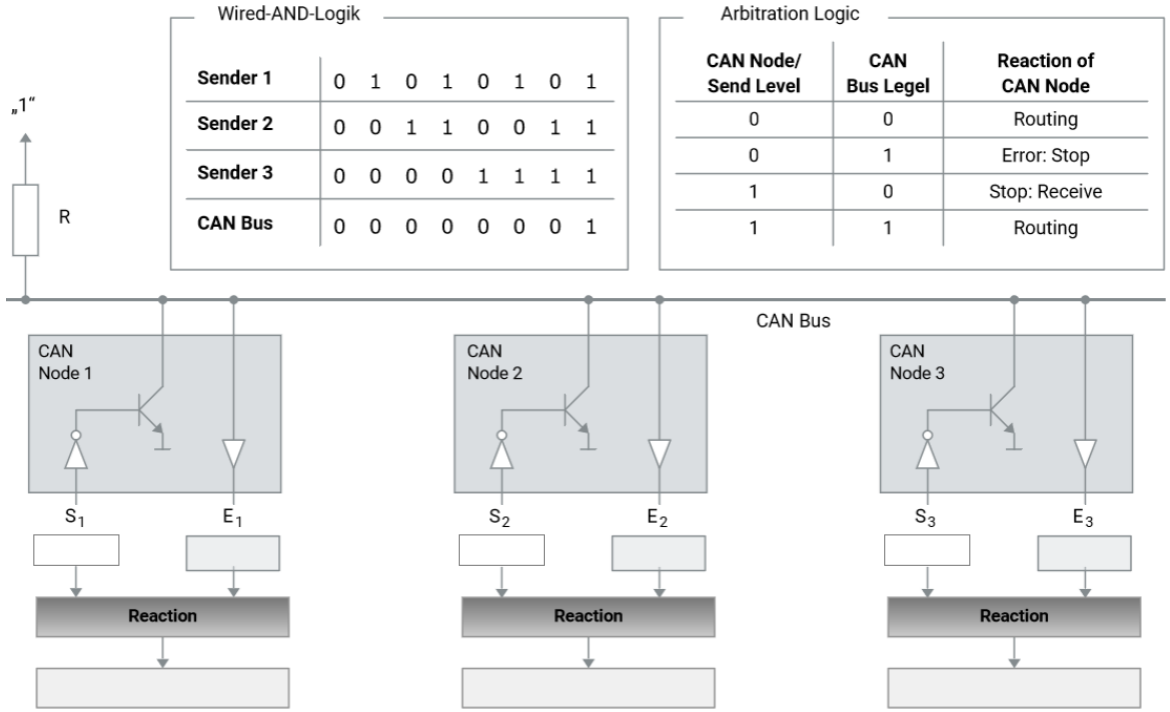




FH JOANNEUM
Electronic Engineering

Arbitration and priorisation

- Bitwise arbitration, wired-AND logic
- To get access to the bus a node needs message with the highest priority
- The smaller the Identifier value the higher the priority



FH JOANNEUM
Electronic Engineering

Arbitration and priorisation

Wired-AND-Logik

Sender 1	0	1	0	1	0	1	0	1
Sender 2	0	0	1	1	0	0	1	1
Sender 3	0	0	0	0	1	1	1	1
CAN Bus	0	0	0	0	0	0	0	1

Arbitration Logic

CAN Node/ Send Level	CAN Bus Legel	Reaction of CAN Node
0	0	Routing
0	1	Error: Stop
1	0	Stop: Receive
1	1	Routing

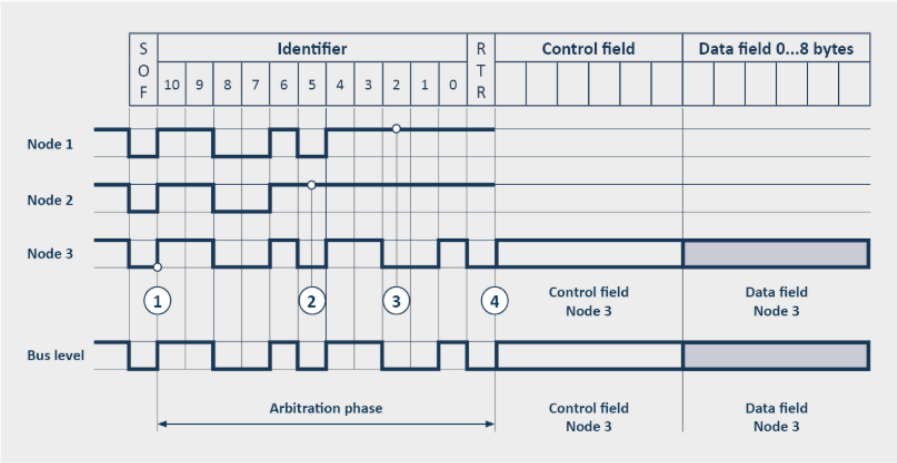
In this example, the bus is at **H level (5 V)** only when none of the nodes pulls the bus line to **L level**.
The H level therefore corresponds to the **recessive level**, while the L level corresponds to the **dominant level**.
As long as the bus is idle, it remains at the recessive level.
A node that starts using the bus indicates this by transmitting a dominant bit (**Start of Frame, SOF**).

During the arbitration phase, each transmitting node compares the bus level it has transmitted with the level actually present on the bus.
Any node that has transmitted a recessive bit but detects a dominant bit immediately stops its arbitration attempt and becomes a receiver of the message currently being transmitted by another node.
Provided that each identifier value is assigned to only one message and that a logical zero is represented by a dominant bus level, only the node whose message has the **lowest identifier value** remains in control of the bus at the end of the arbitration process.

Therefore, the **lower the identifier value assigned to a message, the higher the priority of that message**.

FH JOANNEUM
Electronic Engineering

Arbitration and priorisation



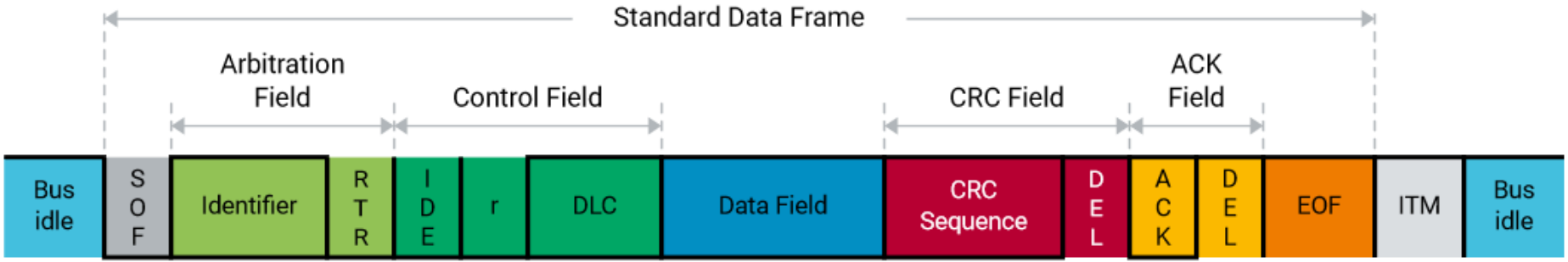
Example of an arbitration process in the CAN protocol.
Nodes 1, 2, and 3 start an arbitration attempt simultaneously (1).
Node 2 loses the right to access the bus at point (2), and Node 1 at point (3).
Both nodes therefore switch to the receive state.
At the end of the CAN bus arbitration phase (4), only Node 3 retains the right to access the bus and transmits its message onto the bus.

CAN Message Format

- Data Frame
  - Transmitted message contains data (e.g. current engine speed) that is provided by transmitting station (data source)
- Remote Frame
  - Stations can call-in data they need from the data source
  - Example: windshield wiper requests how wet windshield is from rain sensor
- Error Frame
  - If station detects a fault or error
- Overload Frame
  - create a delay between preceding and subsequent data or remote frame

CAN Message Format

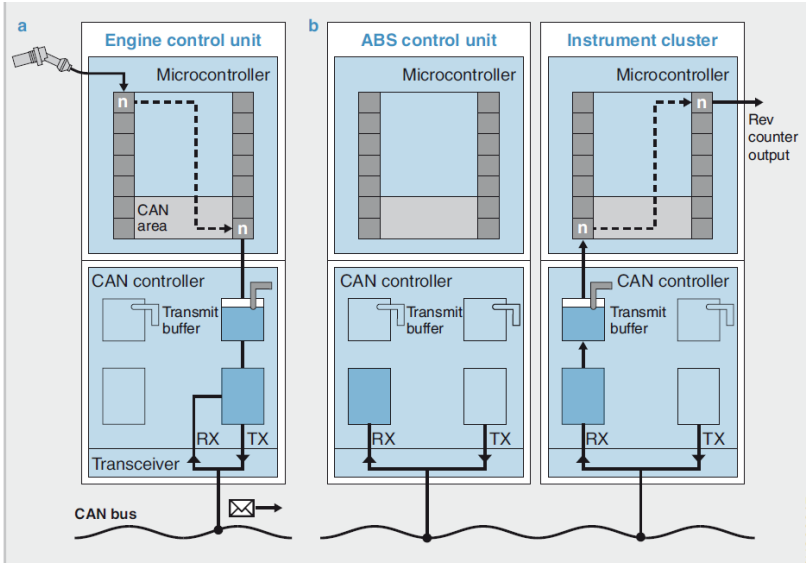
- SOF → start of frame represented by a dominant bit
- Arbitration field → 11-bit or 29-bit (extended) identifier and remote transmission request
- Control field → Identifier extension and number of data length code (DLC) in data field
- Data field → Actual message information 0…8 byte
- CRC field → 15-bit cyclic redundancy checksum (Generator polynom G(x))
- ACK field → Divison frame/polynom →acknowledge receipt by the **receiver**
- EOF → End of frame with 7 recessive bits
- ITM → Interframe space, separate successive messages



FH JOANNEUM
Electronic Engineering

Data transfer sequence

- Engine ECU calculates engine speed from sensor on microcontroller level
- CAN Controller compiles CAN frame and generates bitstream if bus is free
- Bus availability check via CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)
- Tranceiver generates electrical signal and puts it on the bus
- Signal is received by all stations
- CAN controller of receiving node checks message for errors and acceptance, reject or receives message

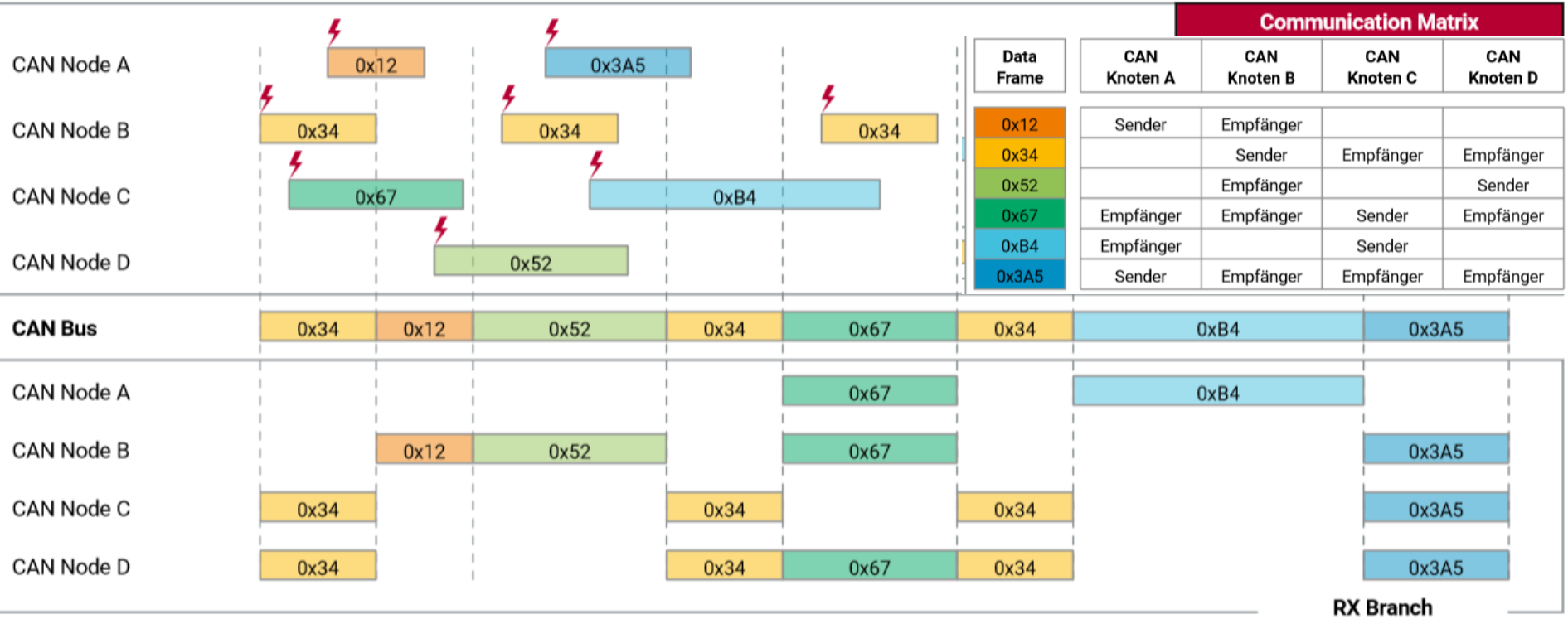


All transmitter/receiver relationships including the meaning of the messages are described in the **communication matrix**

https://elearning.vector.com/mod/page/view.php?id=343

FH JOANNEUM
Electronic Engineering

Exercise: Typical CAN-Communication



Communication Matrix

Data Frame	CAN Knoten A	CAN Knoten B	CAN Knoten C	CAN Knoten D
0x12	Sender	Empfänger		
0x34		Sender	Empfänger	Empfänger
0x52		Empfänger		Sender
0x67	Empfänger	Empfänger	Sender	Empfänger
0xB4	Empfänger		Sender	
0x3A5	Sender	Empfänger	Empfänger	Empfänger

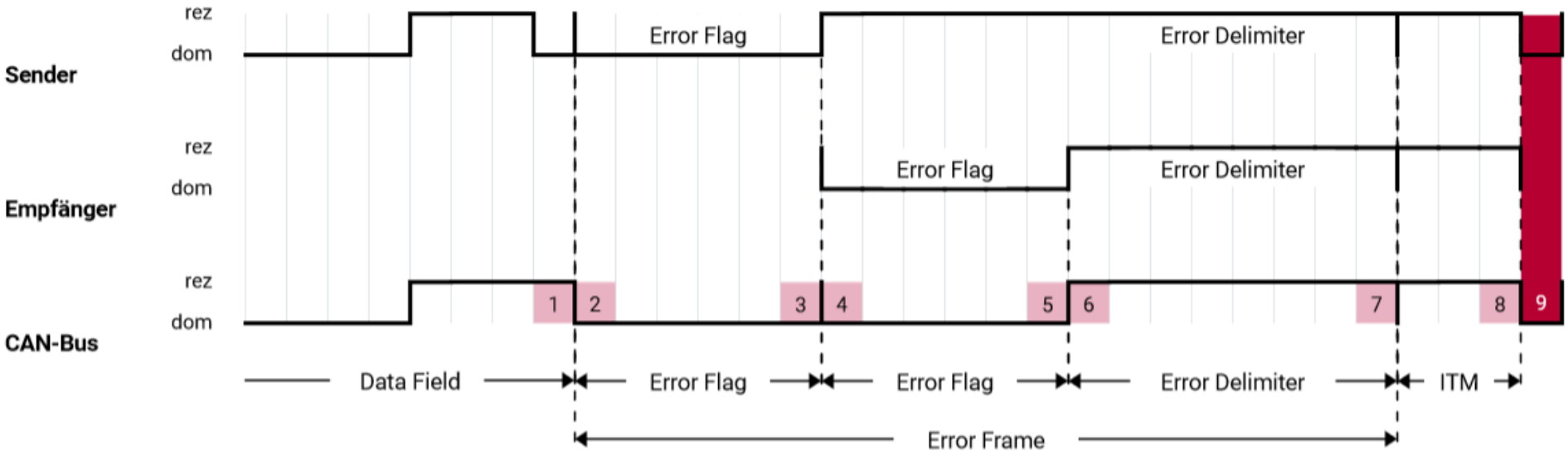
CAN Error processing

- If an error is detected by a node , message transfer is aborted
- All nodes have to be informed by the node who has detected the error
- Error flag is put on the bus by six dominant bits
- This is a violation of the bitstuffing rule
  - Bitstuffing rule: after five bits of same kind transmission of one complementary bit
- Violation of bitstuffing rule causes error flag at all other nodes
- After error flag, delimiter and intermission data frame is send again (if allowed by arbitration)
- Error counters detect recurring errors, CAN-Controller will be cut from bus at counter overload



Error processing

- 1-2 Bus monitoring error detected by sender, error flag set and bit stuffing rule violated
- 3-4 Receiver detects bitstuffing rule violation and stets error flag, sender in delimiter state
- 5-6 Receiver has finished error flag, both sender and receiver go in delimiter mode
- 7-9 Delimiter and intermission finished, data frame starts again with SOF



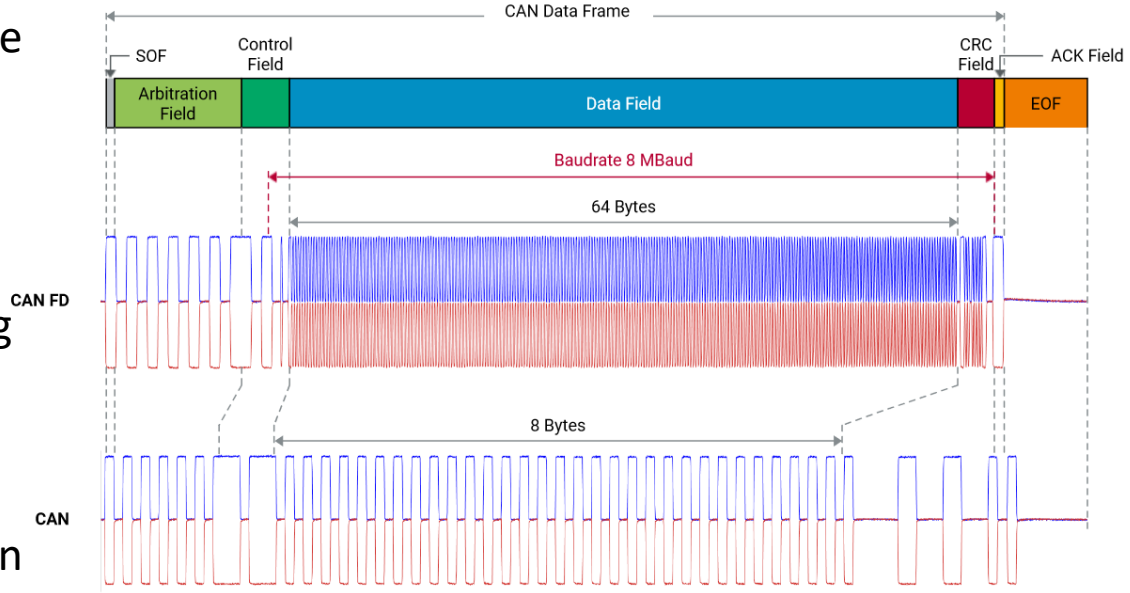
FH JOANNEUM
Electronic Engineering

CAN-FD

FH JOANNEUM
Electronic Engineering

CAN FD Motivation and principle

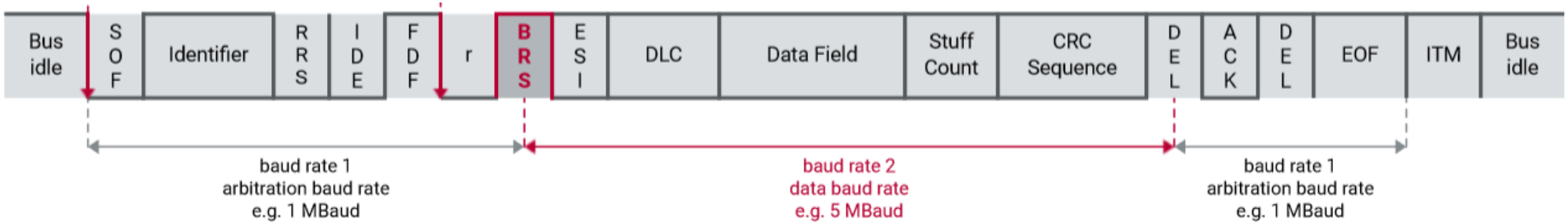
- Bandwidth requirements increase due to additional vehicle functions
- General exchange CAN by other bus technologies with higher bandwidth not suitable regarding costs and development efforts
- Bandwidth limitating factor for CAN is parallel node communication during arbitration (and acknowledge ACK) period
- Idea: Increase the bandwidth only during data field



FH JOANNEUM
Electronic Engineering

CAN FD Motivation and principle

- CAN Reserve bit is used as CAN FD indicator
- FDF (Flexible data rate format) indicates transmission of CAN or CAN FD frame
- Bit rate switch (BRS) announces increase of baud rate
- Switch back to baudrate 1 at CRC delimiter
- Backwards compatibility to CAN, but no upwards compatibility



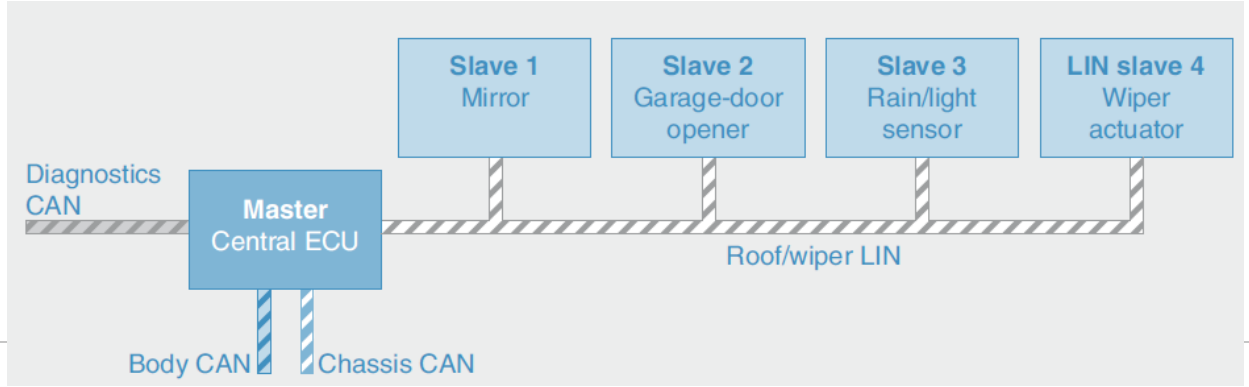
FH JOANNEUM
Electronic Engineering

LIN-BUS

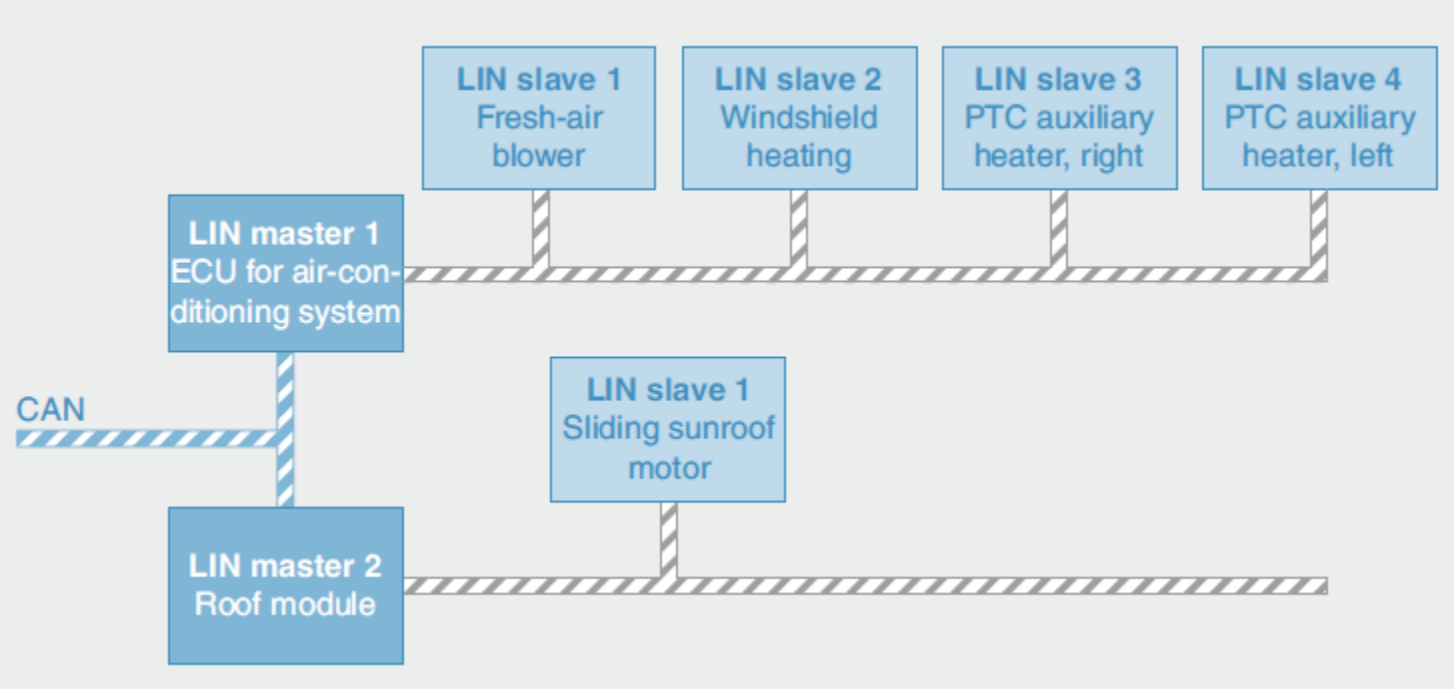
FH | JOANNEUM
Electronic Engineering

LIN (local interconnect network) – Bus

- Low-cost unshielded single-wire bus system
- Low data rates max 20kbit/sec and limited to 16 bus subscribers
- Local subsystem within demarcated installation space (e.g. door)
- Master/slave topology with connection to superordinate CAN
- Time-synchronous communication where master defines time grid
- Mainly comfort applications: Door modules, sunroof, seat adjustment etc.



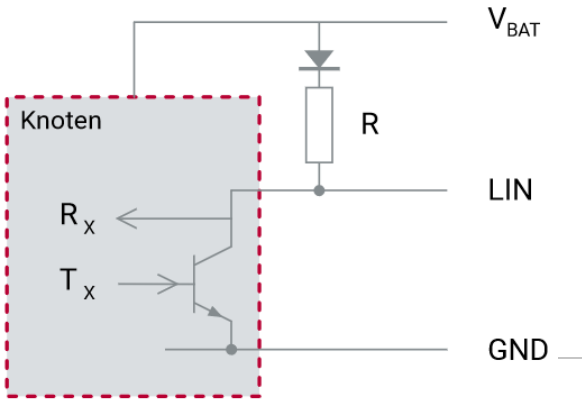
LIN networking example – AC control

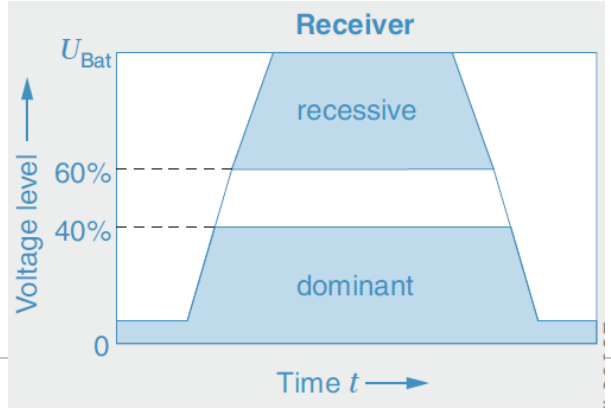


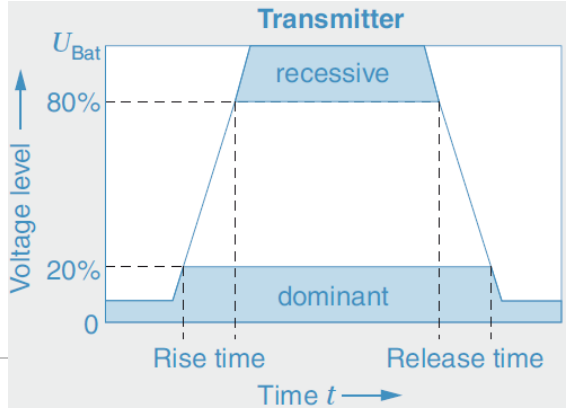
FH JOANNEUM
Electronic Engineering

LIN data transmission system

- Open collector circuit, wired-AND logic same as CAN
- Dominant level corresponds to ~ 0V (vehicle ground) and represents logical 0
- Recessive level corresponds to VBAT and represents logical 1
  - Pull-up Resistance 1kΩ (master) and 30kΩ (slaves)
- Stable data transfer by means of tolerance zones for transmitter and receiver
- Data rate limited to 20kbit/sec

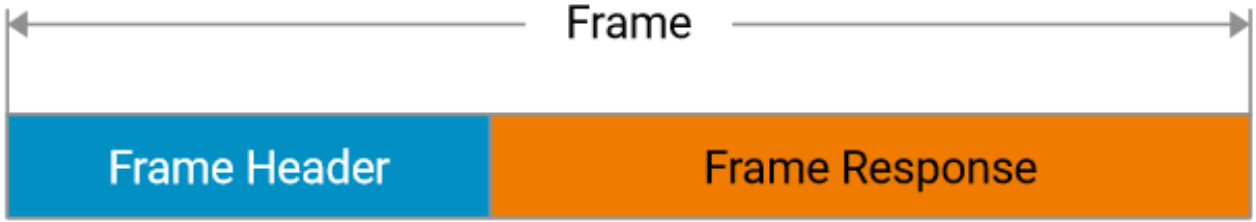






LIN bus access – message format

- Master initiates slave(s) response with frame header
- Slaves answer with frame response
- Deterministic„Delegated token“ bus access → No collisions or arbitration
- Predictable data transfer, defined **schedule**
- LIN configuration in *.ldf-file
- LIN standard describes several types of frames
- Several types of frame, unconditional frame described here, unique header and response



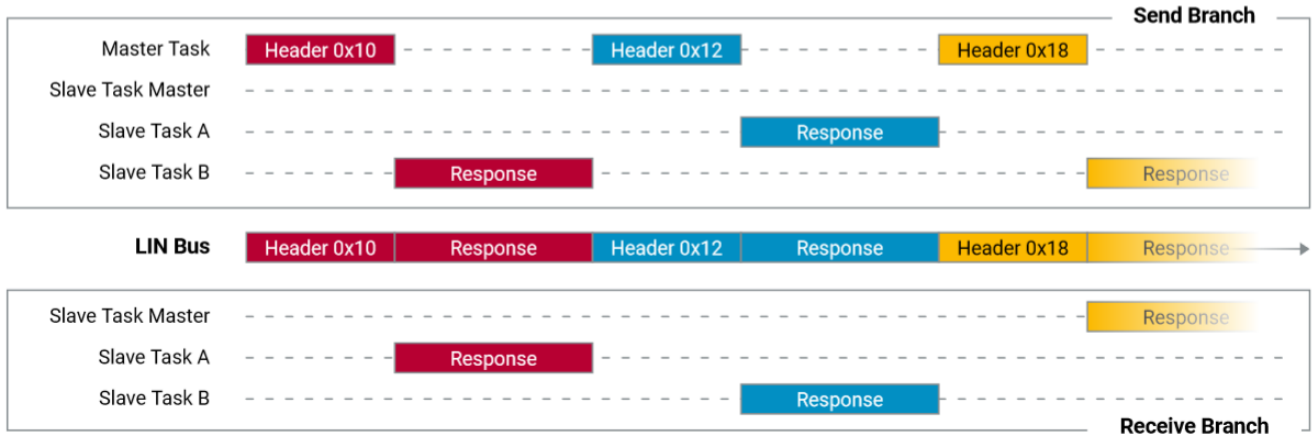
FH JOANNEUM
Electronic Engineering

LIN Schedule

Communication Matrix

Schedule		Frame \ Slave Task	Slave Task Master	Slave Task A	Slave Task B
T_1	Frame Slot 1	Unconditional Frame ID=0x10		Receiver	Sender
T_2	Frame Slot 2	Unconditional Frame ID=0x12		Sender	Receiver
T_3	Frame Slot 3	Unconditional Frame ID=0x18	Receiver		Sender
T_4	Frame Slot 4	Unconditional Frame ID=0x1C	Receiver	Sender	Receiver
T_5	Frame Slot 5	Unconditional Frame ID=0x20	Receiver		Sender
T_6	Frame Slot 6	Unconditional Frame ID=0x24	Sender	Receiver	

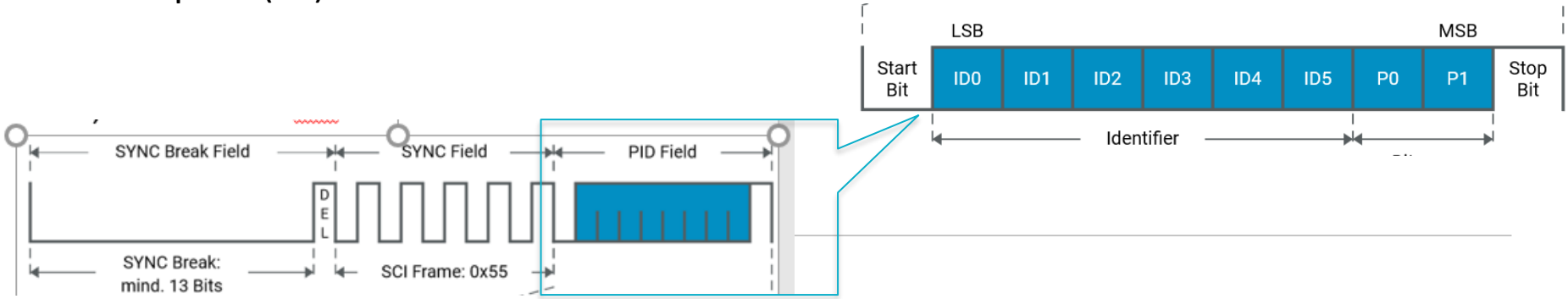
Communication Cycle



FH JOANNEUM
Electronic Engineering

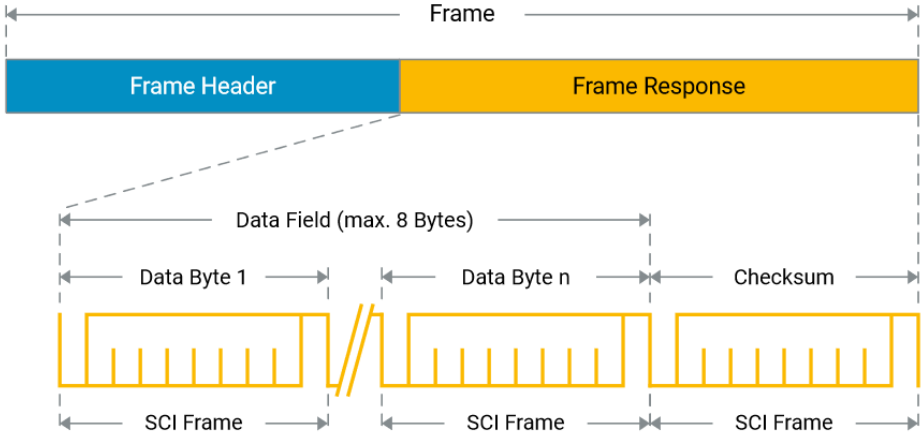
LIN Frame header

- Master task
- Sync break field initiates begin of the header
- Sync Field includes clock frequency and is used for synchronization with slaves
- Protected identifier (PID) field
  - Unique identifier for slave communication, defined in *ldf
  - 6-bit identifier + 2 parity bits
- Between header and response frame there is a pause time called response space (RS)



LIN Frame response

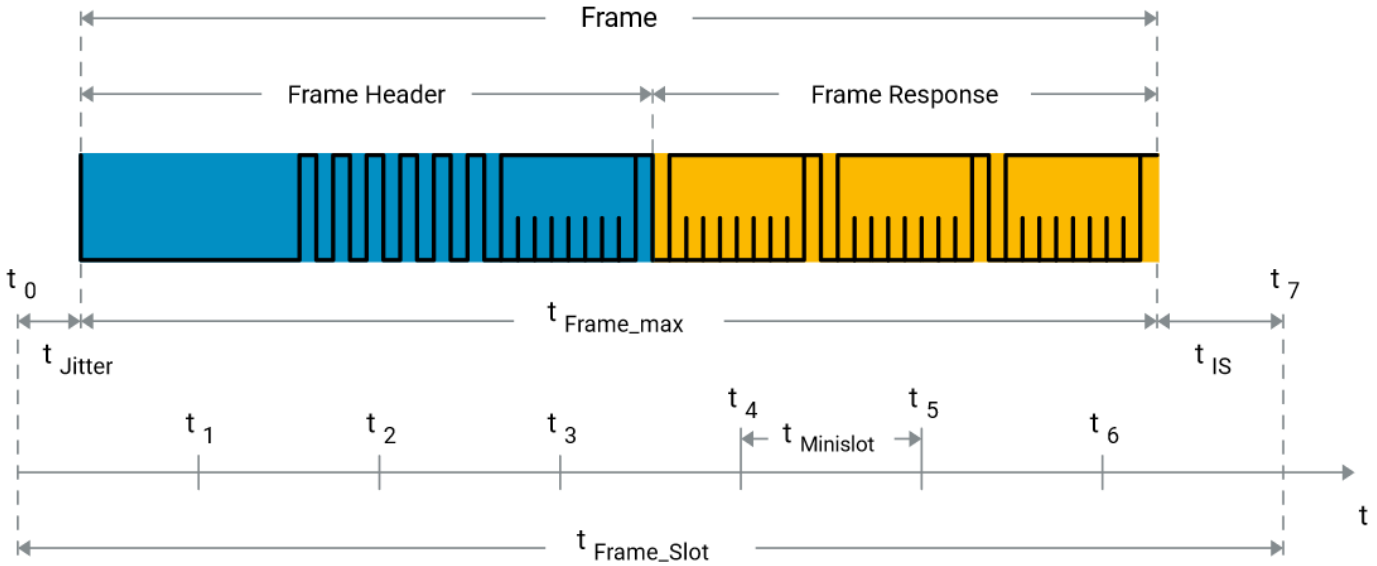
- Slave task
- Slaves answer with max. 8 byte response to the frame header (data field)
- Also slave task of the master can be accessed
- Bytes are transferred from LSB to MSB



FH JOANNEUM
Electronic Engineering

LIN Frame – time base

- The LIN schedule is organized in time slots to transfer one frame
- Size of the slots is defined by minislots as communication time base
- Time reserve to ensure communication (Jitter before and Interframe Space after frame) up to 40%



LIN Error Detection

- Bitmonitoring (sender task)
  - Compare each bit on the bus with bus level
- Checksum (Receiver task)
  - Check of incoming data bytes w/o PID (classic/enhanced checksum)
  - Sum of received data and received checksum has to 0xFF
- Parity check (receiver task)
  - Check of P0 and P1
- Slave responding check (receiver task)
  - Check whether frame response after header is transferred
- Sync field check (receiver task)
  - Check whether master sync rate is in between tolerance range
- Reaction to errors part of individual design, not described in the standard

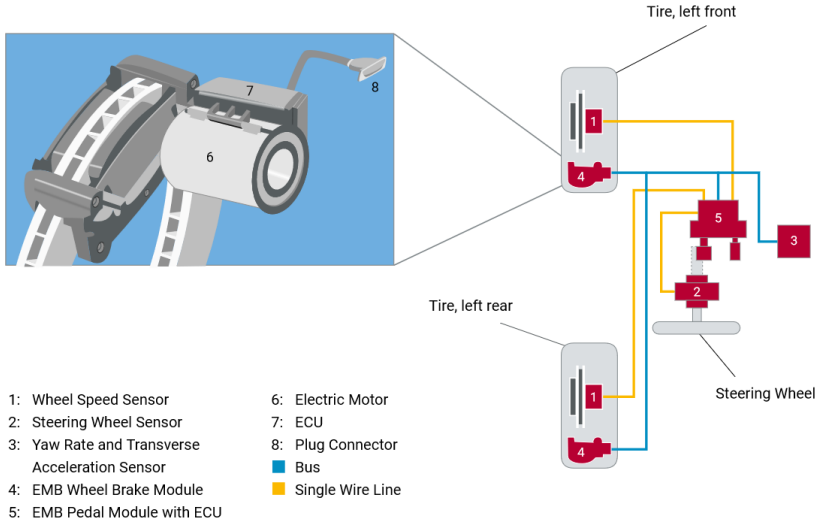
FH JOANNEUM
Electronic Engineering

FlexRay

FH JOANNEUM
Electronic Engineering

FlexRay Introduction

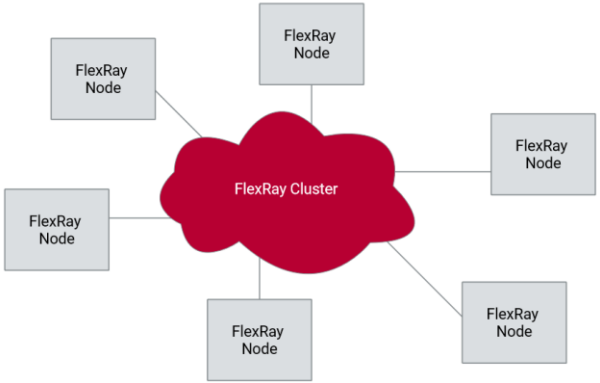
- Developed by BMW and DaimlerChrysler for safety-critical applications (e.g. steer-by-wire)
- Described ISO 17458
- Real-time communication systems with a time-triggered approach
- High transfer-rates up to 20Mbit/s, deterministic and fault-tolerant, while being flexible
- Time-triggered communication architecture (TDMA)

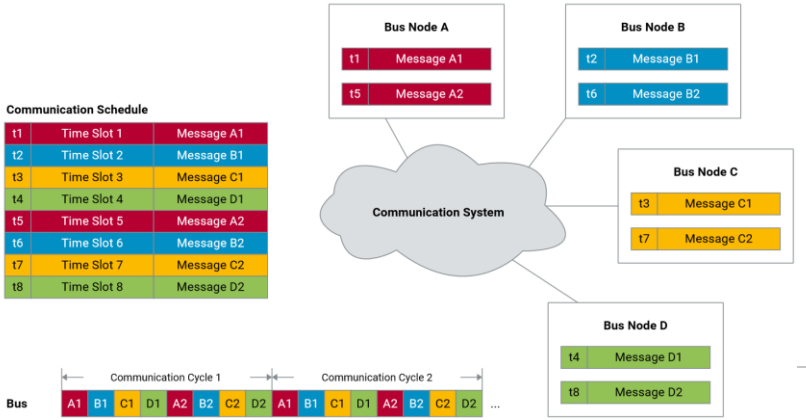


1: Wheel Speed Sensor
2: Steering Wheel Sensor
3: Yaw Rate and Transverse Acceleration Sensor
4: EMB Wheel Brake Module
5: EMB Pedal Module with ECU
6: Electric Motor
7: ECU
8: Plug Connector
Bus
Single Wire Line

FlexRay Architecture

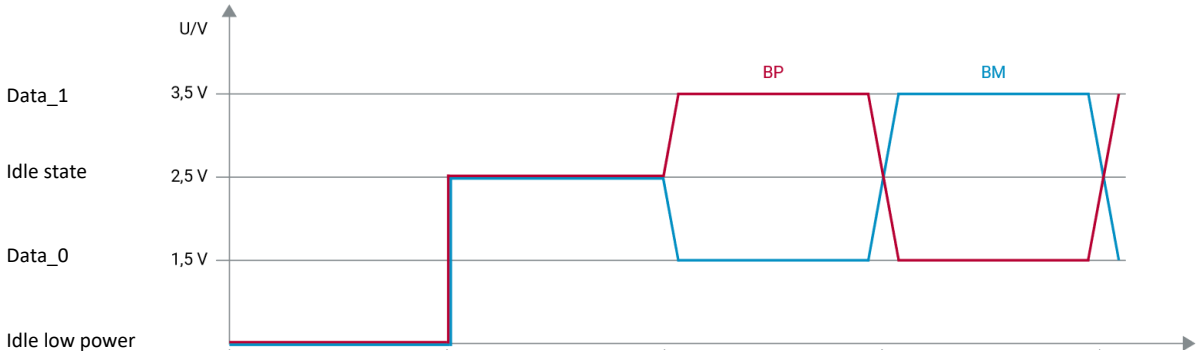
- A FlexRay communication system (Cluster) consists of nodes and transmission medium (bus)
- Two transmission channels (redundancy 10Mbit/s or increase of bandwidth 20Mbit/s individually per message)
- Several bus topologies are possible (point-to-point, bus, star, hybrid,…)
- Twisted-pair cabling design shielded or unshielded
- TDMA principle: Nodes must conform to communication schedule with specific time slot to each message per communication cycle, send times of all FlexRay messages prescribed





FlexRay Bus Level

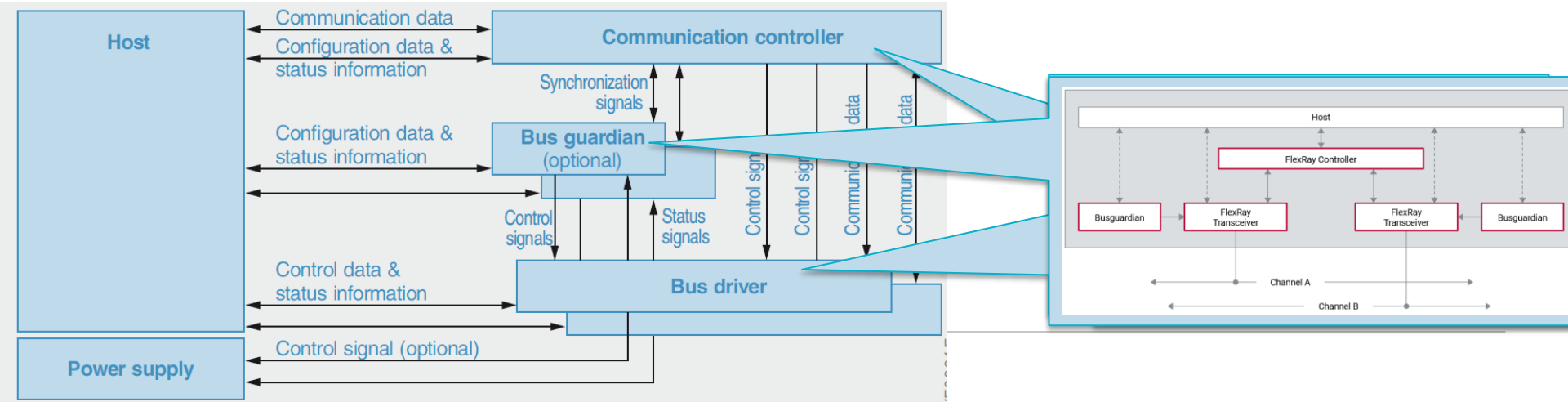
- Non-return-to zero (NRZ) method
- Transmission of voltage differences , FlexRay bus consists of the two lines Bus Plus (BP) and Bus Minus (BM)
- Dominant: differential voltage ≠ 0V
- Recessive: differential voltage ~ 0V
- FlexRay defines four bus states
  - Idle_LP (low power) recessive → 0V +-200mV , identifies start of transmission (Bus sleep)
  - Idle state recessive → 2.5V at BP and BM
  - Data_1 dominant → logical „1“ voltage difference 2V
  - Data_0 dominant → logical „0“ voltage difference -2V



FH JOANNEUM
Electronic Engineering

FlexRay Hardware

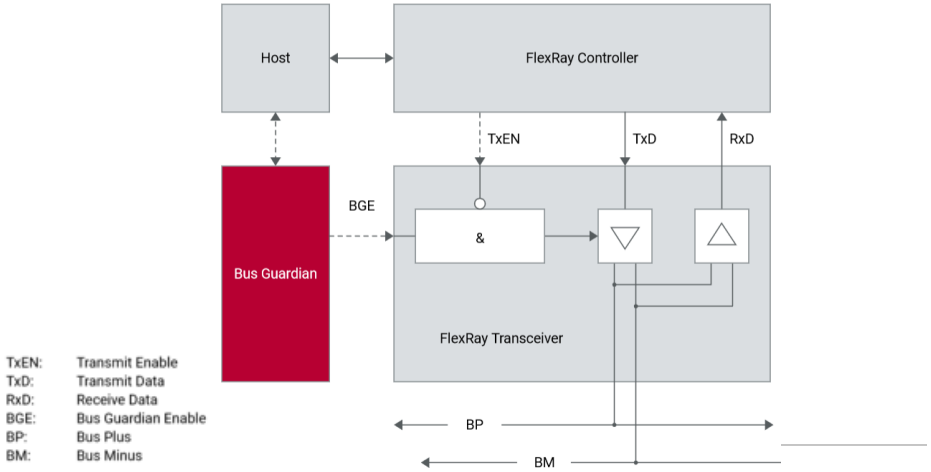
- A node comprises of
  - Host processor: gathers sensor information
  - Communication controller
    - Controller realizes FlexRay protocol , Transceiver creates bitstream for each channel
  - Bus driver: Converts bitstream into voltage signals and vice versa (Transceiver), power management
  - Bus guardian (optional): Checks and authorizes transmission , local inside one node or central for all nodes

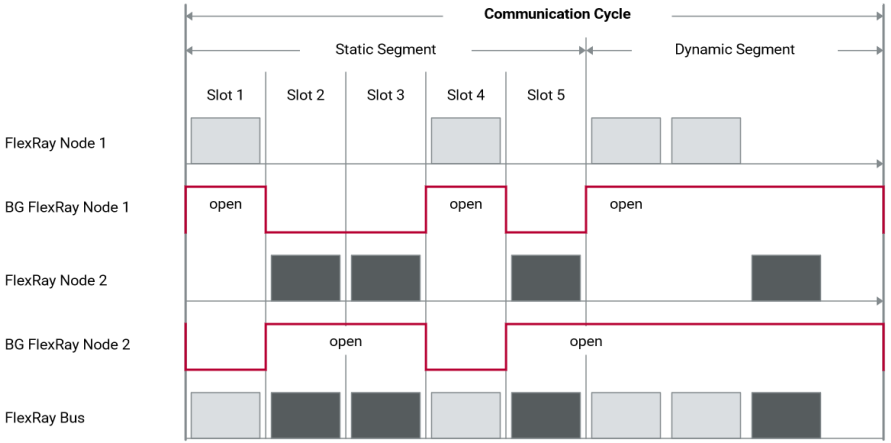


FH JOANNEUM
Electronic Engineering

Bus Guardian

- Guardians prevent unauthorized transmissions due to unsynchronized FlexRay nodes
- Each static slot is authorized by the guardian
- Authorization for dynamic segment can only be done completely
- Guardians ideally have their own time base to check communication schedule and time in the FlexRay cluster



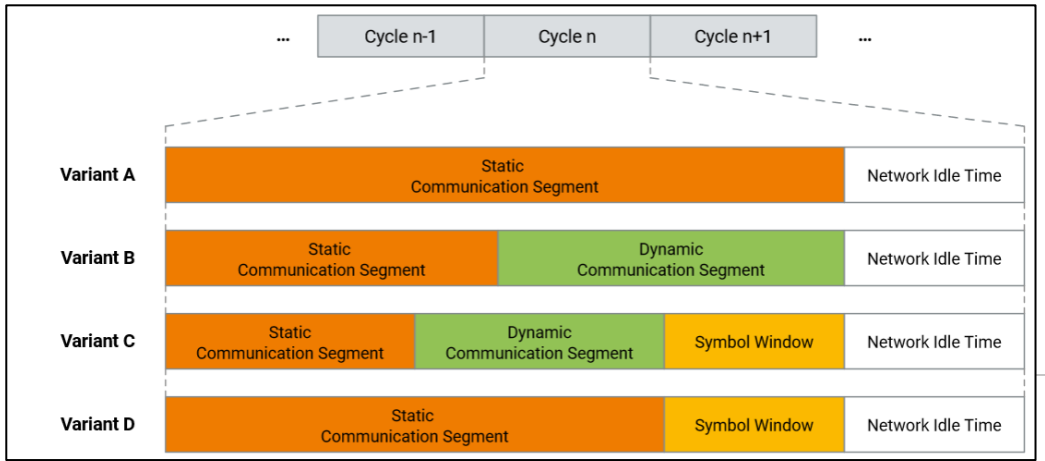


Assigned to FlexRay node 1 are slots 1 and 4 on channel A, and assigned to FlexRay node 2 are slots 2, 3 and 5 on the same channel. The two bus guardians (BG) monitor the relevant slots. In the dynamic segment, both BGs open up the path to the bus.

FH JOANNEUM
Electronic Engineering

Communication Cycle

- Data communication in a FlexRay cluster is periodical and is based on a communication schedule
- Four variants of communication cycle possible
- At least two segments, **static segment** and the network idle time (NIT) segment
- NIT segment is needed to synchronize the local clocks, cycle terminates
- Optional **dynamic segment**: event-driven, asynchronous message transmission; if included it follows the static segment
- Optional Symbol window: Collision avoidance symbol, media test symbol or wake-up symbol



Communication Schedule

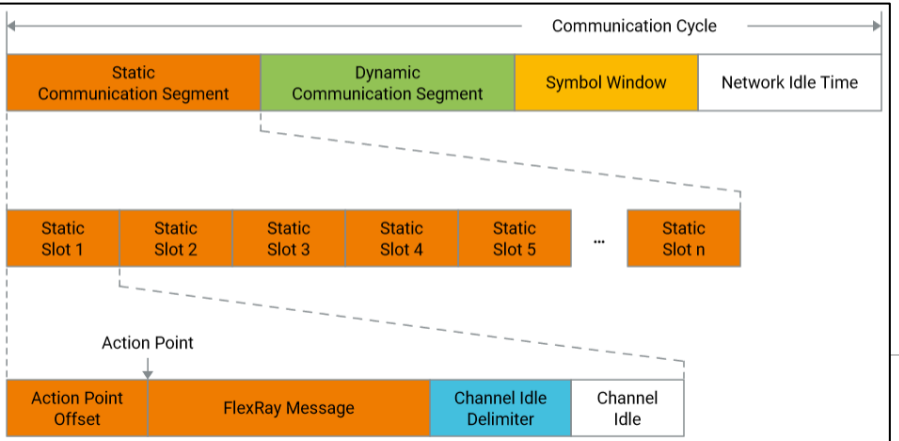
	Slot	Start	Node	Message
Static Segment	1	t_1	A	A1
	2	t_2	B	B1
	3	t_3	D	D1
	4	t_4	E	E1
	5	t_5	C	C3
Dynamic Segment				

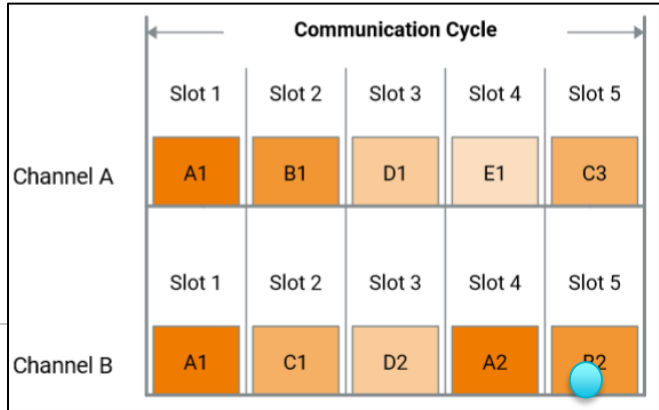
FH JOANNEUM
Electronic Engineering

Static segment

- FTDMA method (Flexible Time Division Multiple Access)
- Fixed number of transmission windows „static slots“ with fixed equal length
- 2-1023 static slots possible, exactly one message per static slot
- Message sent simultaneously on one channel (**bandwidth increase**) or two channels (**redundancy increase**) according communication schedule
- In each slot the exclusive frame with corresponding ID is sent periodically
- Action point offset and channel idle to synchronize to synchronize advancing and delayed receiver clocks

Slot	Node	Message	Channel
1	Node A	A1	A
		A1	B
2	Node B	B1	A
	Node C	C1	B
3	Node D	D1	A
		D2	B
4	Node E	E1	A
	Node A	A2	B
5	Node C	C3	A
	Node B	B2	B





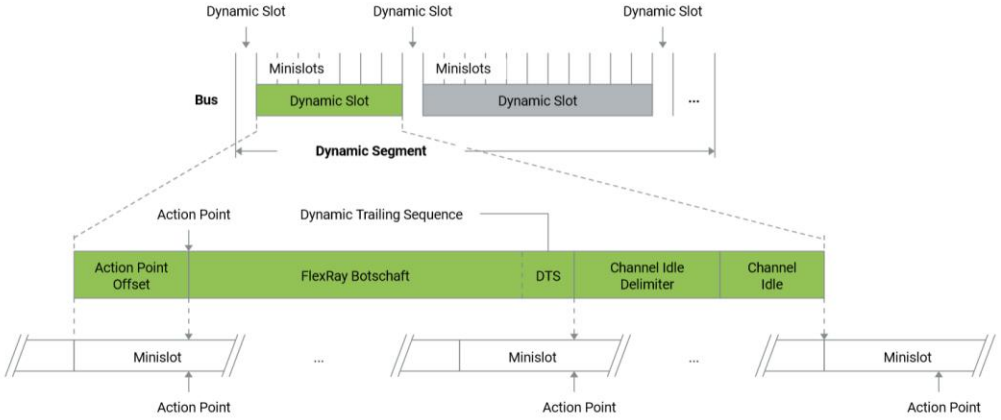
FH JOANNEUM
Electronic Engineering

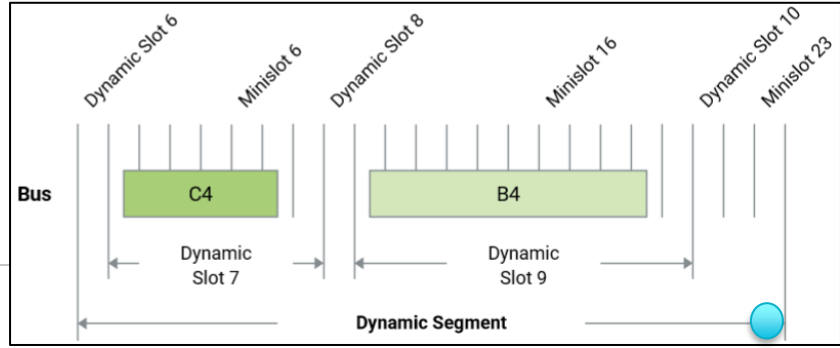
Dynamic Segment

- FTDMA (Flexible Time Division Multiple access)
- Dynamic segments also based on schedule, always same length
- Dynamic segment consist of dynamic slots (minimum size 1 minislot)
- Dynamic slot layout similar to static slot, messages have different payload sizes
- Dynamic segment begins with all nodes incrementing their local counters
- If there is a send request, send dynamic message matching the counter
- If there is no send request, increment counter again (length 1 minislot)
- Procedure is repeated until the dynamic segment is no longer long to transmit a dynamic message

Communication Schedule (Dynamic Segment)

Slot	Node	Message	Event
6	A	A4	
7	C	C4	⚡
8	D	D3	
9	B	B4	⚡
10	E	E3	

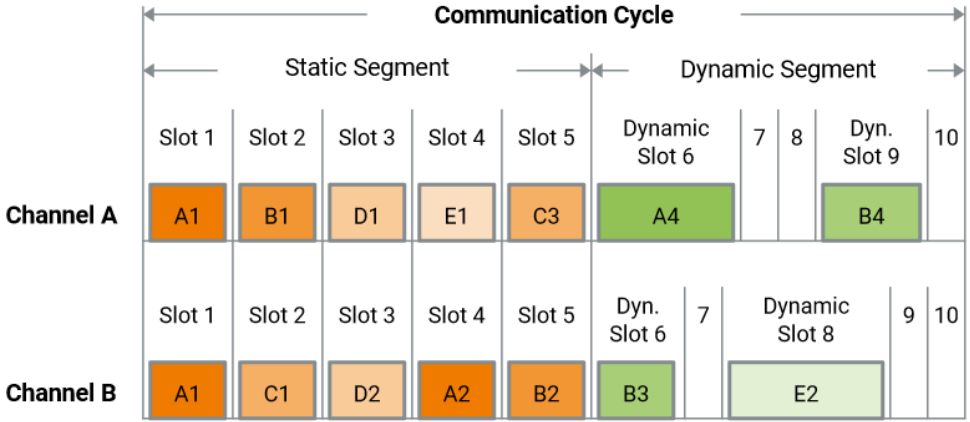




FH JOANNEUM
Electronic Engineering

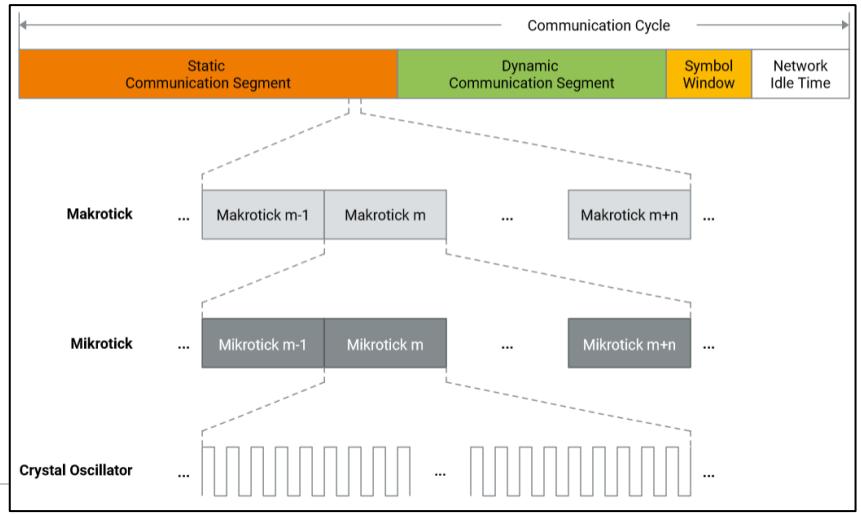
Exercise: FlexRay Communication cycle example

	Slot	Node	Frame	Channel	Event
Static Segment	1	Node A	A1	A	
			A1	B	
	2	Node B	B1	A	
		Node C	C1	B	
	3	Node D	D1	A	
			D2	B	
	4	Node E	E1	A	
		Node A	A2	B	
	5	Node C	C3	A	
		Node B	B2	B	
Dynamic Segment	6	Node A	A4	A	⚡
		Node B	B3	B	⚡
	7	Node C	C4	A	
		Node D	D3	B	
	8	Node D	D3	A	
		Node E	E2	B	⚡
	9	Node B	B4	A	⚡
		Node A	A5	B	
	10	Node E	E3	A	
		Node C	C5	B	



FlexRay Clock Control

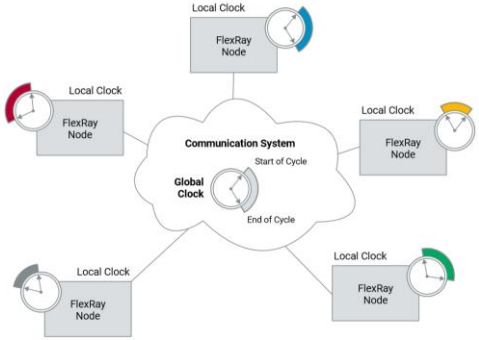
- Each network node has its own internal clock
- System-wide global time preset is necessary for all nodes because bus access is controlled by means of time slots
- Microtick level
  - Bottom-level, not a system-wide variable
  - Derived directly from the oscillator clock of the network node
- Macrotick level
  - Range in which exactly one frame can be sent
  - Lowermost system-wide unit
  - #microticks/macrotick not the same for all macroticks
- Communication cycle level
  - Top-level
  - Contains the same, fixed number of macroticks

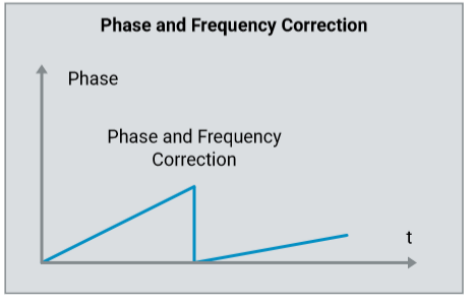


FH | JOANNEUM
Electronic Engineering

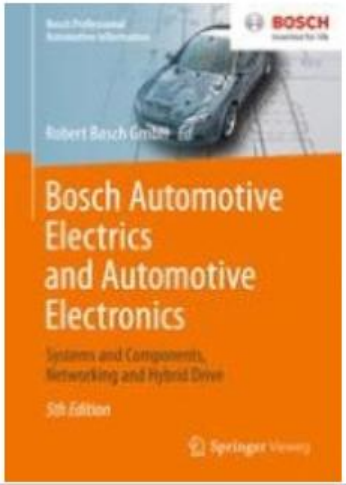
FlexRay Cluster Synchronization

- All FlexRay nodes share time consensus, because all activities are triggered in the time sequence
- All communication cycles and static slots always have to begin at the same point and are of equal length
- Time consensus can only be produced cooperatively based on the local time bases
- Tolerances produce different frequencies and phases at identical nominal frequencies
- Correct local clocks so that all local clocks in the FlexRay cluster run synchronous to a global clock up to a defined deviation → phase and frequency correction
- Some FlexRay nodes act as sync nodes which transmit synch messages (sync frame indicator set)
- All other nodes compare the a priori known time points with the points at which the sync messages actually arrive and compute their phase and frequency correction value using the fault tolerant midpoint (FTM) algorithm





References and quotations



Buchreihe: Bosch Professional Automotive Information

Herausgeber: Robert Bosch GmbH

Verlag: Springer Fachmedien Wiesbaden

Print ISBN: 978-3-658-01783-5
Electronic ISBN: 978-3-658-01784-2

Enthalten in: Springer Professional "Wirtschaft+Technik" , Springer Professional "Technik"

https://elearning.vector.com
https://www.eecs.umich.edu/courses/eecs461/doc/CAN_notes.pdf
https://www.uni-kassel.de/eecs/fileadmin/datas/fb16/Fachgebiete/FSG/Download/Lehre/PFS/
http://www.lin-subbus.org